Dermatol Ther (Heidelb)
https://doi.org/10.1007/s13555-026-01677-2

REVIEW

Artificial Intelligence in Hair Disorders: A Narrative Review

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Received: January 7, 2026 / Accepted: January 28, 2026
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ABSTRACT

Artificial intelligence (AI) is increasingly being incorporated into dermatology, with hair disorders representing a particularly suitable field for its application. The assessment and management of alopecias rely heavily on visual evaluation, pattern recognition, longitudinal monitoring, and integration of heterogeneous clinical and biological data; processes that are often limited by interobserver variability and incomplete standardization. In recent years, AI-based approaches have been applied across multiple domains in the clinical assessment of hair disorders, including objective quantitative assessment, diagnostic support, personalized medicine, drug development, robotic-assisted procedures, and analysis of patient-reported outcomes and clinical communication. Among these applications, tools enabling automated and reproducible measurement of hair parameters are currently the most established in clinical practice, whereas diagnostic and predictive models remain largely investigational. This narrative review summarizes current and emerging applications of artificial intelligence in hair disorders, with a focus on their clinical relevance and practical implementation. In addition, methodological limitations, ethical considerations, and challenges related to external validation, interpretability, and data bias are discussed. Although AI-based tools are not yet able to replace clinical judgement, they hold significant potential to complement dermatologists' expertise and improve objectivity, efficiency, and personalization in the management of hair disorders.

Keywords: Alopecia; Hair loss; Machine learning; Neural networks; Trichology

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Published online: 27 February 2026

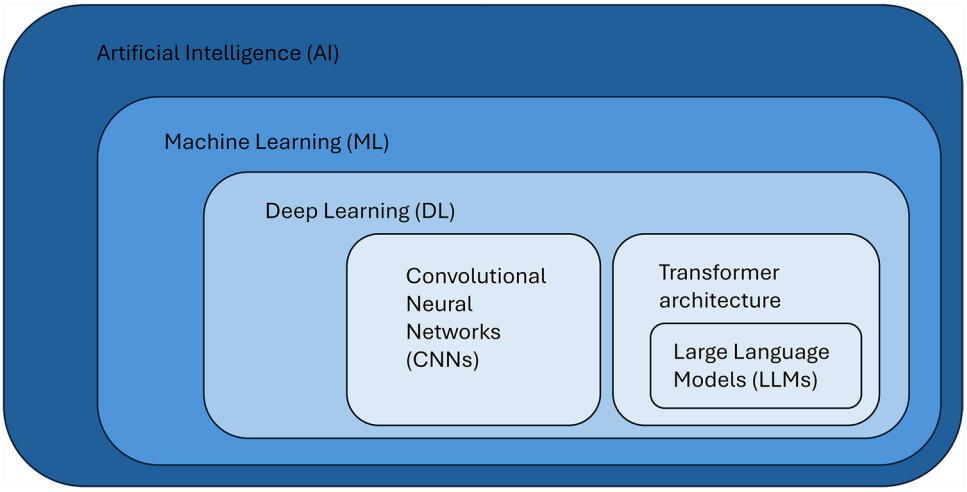

Fig. 1 Conceptual hierarchical organization of artificial intelligence methods applied to medicine, with particular relevance to hair disorders

Key Summary Points

Artificial intelligence (AI) is being increasingly applied across multiple aspects of hair disorders, including objective quantitative assessment, diagnostic support, personalized medicine, drug development, robotics, and patient-centered communication.

Among current applications, AI-based tools enabling automated and reproducible measurement of hair parameters represent the most established and clinically mature use cases, whereas diagnostic and predictive models remain largely investigational.

Emerging AI approaches show potential to improve biomarker discovery, treatment stratification, and patient education in hair disorders, although robust clinical validation is still required.

While AI-based technologies are not intended to replace clinical expertise, they may support dermatologists by improving objectivity, reproducibility, and scalability in both clinical practice and research settings.

INTRODUCTION

Artificial intelligence (AI) has emerged as an increasingly relevant tool in medicine, driven by advances in computational power, data availability, and algorithmic development. In dermatology, and particularly in the field of hair disorders, AI-based approaches have gained attention owing to the visual nature of clinical assessment, the growing use of digital imaging techniques, and the availability of large, heterogeneous datasets derived from clinical, trichoscopic, histopathological, and molecular sources.

The term "artificial intelligence" encompasses a broad and evolving set of computational methods, ranging from traditional machine learning (ML) algorithms to advanced deep learning (DL) architectures and large language models (LLMs). These concepts are often used interchangeably in the medical literature, which may lead to ambiguity and misinterpretation among clinicians. For clarity, a conceptual hierarchical organization of AI and related learning approaches is provided (Fig. 1), together with concise definitions of the main AI-related terms (Table 1).

In recent years, AI applications in hair disorders have progressively expanded to support multiple aspects of clinical practice and research, including

Table 1 Definitions of artificial intelligence concepts applied to medicine, with particular relevance to hair disorders

Concept	Definition
Artificial Intelligence (AI)	A broad field of computer science focused on the development of systems capable of performing tasks that normally require human intelligence, including reasoning, learning, perception, and decision-making. In medicine, AI supports diagnosis, prognosis, and clinical decision-making [45, 51].
Machine Learning (ML)	A subset of artificial intelligence that enables systems to learn patterns and relationships from data without being explicitly programmed. ML includes a wide range of algorithms, from classical statistical models to neural network-based approaches [52, 53].
Deep Learning (DL)	A subset of machine learning that relies on artificial neural networks with multiple hidden layers (deep neural networks) to learn complex patterns from data. DL is particularly effective for analyzing data such as medical images, biomedical signals, and language [54].
Convolutional Neural Networks (CNNs)	A class of deep learning architectures specifically designed to process grid-like data, such as images. CNNs use convolutional layers to automatically learn spatial features and are widely applied in medical image analysis, including clinical and dermoscopic/trichoscopic photographs [55, 56].
Transformer Architectures	A family of deep learning architectures based not only on self-attention mechanisms, primarily applied in natural language processing, but also extended to other domains such as multimodal data analysis [57].
Large Language Models (LLMs)	A subclass of transformer-based deep learning models trained on massive text corpora to understand and generate natural language. LLMs are capable of performing a wide range of language-related tasks, including text generation, summarization, question answering, and clinical text analysis [58, 59].

objective disease quantification, diagnosis, biomarker discovery, drug development, and patient-centered communication. Hair disorders constitute a particularly suitable field for AI-based approaches, as their assessment and management rely heavily on visual evaluation, pattern recognition, longitudinal follow-up, and the integration of heterogeneous clinical and biological data; processes often limited by interobserver variability and incomplete standardization.

This narrative review provides an overview of current and emerging applications of artificial intelligence in hair disorders and discusses their potential clinical impact, methodological limitations, and ethical considerations.

MATERIALS AND METHODS

A comprehensive literature review on AI applications in hair disorders was conducted in October 2025 and updated thereafter, with the latest update in January 2026. A PubMed database search was performed using the keywords (“artificial intelligence” OR “machine learning”) AND (“trichology” OR “alopecia” OR “hair loss”).

No date restrictions or additional filters were applied.

The search yielded 156 results. All retrieved records were screened without applying title or abstract keyword filters. The title and abstract of every search result were reviewed, and full-text articles were accessed whenever necessary to determine their eligibility. Relevant works found through cross-reference searches were incorporated and inspected carefully. Studies that did not align with the scope of AI applications in hair disorders were excluded. Full texts of all eligible articles were subsequently examined in detail. The references discussed in the manuscript represent those studies considered relevant to the application of artificial intelligence in hair disorders.

This article is based on previously conducted studies and does not contain any new studies with human participants or animals performed by any of the authors. The extracted information is synthesized in a narrative format, with the aim of highlighting key insights and clinically relevant learnings for dermatologists.

RESULTS AND DISCUSSION

This review has been organized according to the major domains in which AI has been applied to hair disorders, with the aim of enhancing its potential impact on daily clinical practice and understanding. The sections progress from technologies designed to obtain objective quantitative measurements to other applications such as robotics. To facilitate rapid consultation, the studies discussed are collected and organized by primary area of application in Table 2. Among these, objective measurement tools represent the most widely implemented tools in current clinical practice. Nonetheless, none of the available systems has yet demonstrated the capacity to replace the clinical judgement or diagnostic role of the dermatologist.

OBJECTIVE QUANTITATIVE ASSESSMENT

When discussing AI in hair disorders, a variety of concepts emerge; however, the most firmly established applications to date are those enabling objective measurements, such as counting hairs and determining their diameter or length. Multiple systems have been developed for this purpose, most of which rely on trichoscopic imaging [1]. One of the best-known tools is TrichoScan® [2–4], which measures hair density, hair diameter, and hair growth rate. It can also estimate the anagen:telogen ratio, requiring two evaluations approximately 3 days apart. The technique integrates conventional epiluminescence microscopy with automated digital image processing. For accurate assessment, the hair must be trimmed close to the scalp and dyed, and fibers shorter than 5 mm cannot be reliably evaluated. Two additional systems that do not require trimmed hair are Folliscope® [5, 6] and HairMetrix® [7]. Folliscope allows the assessment of hair density and diameter. HairMetrix provides automated measurements of hair density, diameter, and terminal:vellus hair ratio; when the hair is clipped, the system can calculate the anagen:telogen ratio.

These quantitative measurement technologies have predominantly been developed for use in androgenetic alopecia (AGA). On the basis of parameters such as hair density and hair diameter, ML models have been developed to assist in AGA grading [8].

Object detection algorithms have been applied to the automated identification and quantification of hair follicles, most notably using You Only Look Once (YOLO)-based architectures such as YOLOv4 [9] and YOLOv5 [10]. These models enable the detection of individual follicular units and their classification according to the number and type of hairs emerging from each unit [10]. Such approaches have been primarily employed for hair density measurement, particularly in the context of hair transplantation, where objective quantification of follicular units within the occipital donor area is essential

Table 2 Key studies on artificial intelligence applications in hair disorders, organized by clinical domain

Author	Year	Title
Objective quantitative assessment		
Gupta et al. [1]	2021	How good is artificial intelligence (AI) at solving hairy problems? A review of AI applications in hair restoration and hair disorders
Hoffmann [2]	2003	TrichoScan: a novel tool for the analysis of hair growth in vivo
Gassmueller et al. [3]	2009	Validation of TrichoScan® technology as a fully-automated tool for evaluation of hair growth parameters
Kolasinski [4]	2009	A useful method for measuring hair growth parameters and enhancing patient selection for hair restoration surgery
Lee et al. [5]	2012	Assessment of hair density and caliber in Caucasian and Asian female subjects with female pattern hair loss by using the Folliscope
Birnbaum [6]	2018	Evaluation of hair density in different ethnicities in a healthy American population using quantitative trichoscopic analysis
Gao et al. [8]	2022	Deep learning-based trichoscopic image analysis and quantitative model for predicting basic and specific classification in male androgenetic alopecia
Kim et al. [9]	2022	Evaluation of automated measurement of hair density using deep neural networks
Lv et al. [10]	2023	A challenge of deep-learning-based object detection for hair follicle dataset
Urban et al. [11]	2021	Combining deep learning with optical coherence tomography imaging to determine scalp hair and follicle counts
Raymond et al. [12]	2024	Use of artificial intelligence to track platelet-rich plasma treatment outcomes in females with nonscarring alopecia: a case series
Maletic et al. [13]	2022	Autologous platelet-rich plasma (PRP) for treating androgenetic alopecia: a novel treatment protocol standardized on 2 cases
Kuceki et al. [14]	2025	Enhanced hair regrowth with five monthly sessions of minoxidil-dutasteride-copper peptides tattooing for androgenetic alopecia assessed by artificial intelligence and blinded evaluators
Nguyen et al. [15]	2025	Using artificial intelligence to compute Severity of Alopecia Tool scores
Lee et al. [16]	2020	Clinically applicable deep learning framework for measurement of the extent of hair loss in patients with alopecia areata
Diagnosis		
Caro et al. [17]	2025	Analysis of trichoscopic images using deep neural networks for the diagnosis and activity assessment of alopecia areata – A retrospective study
Shakeel et al. [18]	2021	Classification framework for healthy hairs and alopecia areata: a machine learning (ML) approach

Table 2 continued

Author	Year	Title
Personalized medicine		
Wang et al. [19]	2025	Microbial dysbiosis and its diagnostic potential in androgenetic alopecia: insights from multi-kingdom sequencing and machine learning
Xu et al. [20]	2024	Identification of *SLC40A1*, *LCN2*, *CREB5*, and *SLC7A11* as ferroptosis-related biomarkers in alopecia areata through machine learning
McMullen et al. [21]	2024	Applications of machine learning on alopecia areata: a systematic review
Karami et al. [22]	2022	*CHAC1* as a novel biomarker for distinguishing alopecia from other dermatological diseases and determining its severity
Chen et al. [23]	2025	Early prediction of alopecia areata using machine learning modeling of neuro stress immune signatures from multi datasets
Zhang et al. [24]	2021	Prediction of the risk of alopecia areata progressing to alopecia totalis and alopecia universalis: biomarker development with bioinformatics analysis and machine learning
Xiong et al. [25]	2023	Construction of regulatory network for alopecia areata progression and identification of immune monitoring genes based on multiple machine-learning algorithms
Chen et al. [26]	2021	Regulatory network analysis defines unique drug mechanisms of action and facilitates patient-drug matching in alopecia areata clinical trials
Deng et al. [27]	2025	Exosome therapeutics: a paradigm shift in skin repair through multidimensional immunomodulation and biomaterial-driven delivery
Tuan et al. [28]	2021	Prediction of therapeutic outcomes of female pattern hair loss patients based on clinical features with application of artificial intelligence
Alarcón-Soldevilla et al. [29]	2021	Use of artificial intelligence as a predictor of the response to treatment in alopecia areata
Seetan et al. [30]	2026	Machine learning predicts scarring progression in lichen planopilaris: a multidimensional model integrating trichoscopy, vitamin D, and diagnostic timelines
Bhardwaj et al. [31]	2025	Artificial intelligence-based personalization of treatment regimen for hair loss: a 6-month clinical trial
Drug development		
Isswanich et al. [32]	2025	Natural product-inspired bis(trifluoromethyl) phenyl hydroxycinnamate derivatives as promising nonsteroidal inhibitors of human steroid 5α-reductase type-1: synthesis, in vitro, and in silico studies
Ma et al. [33]	2024	A top-down design approach for generating a peptide PROTAC drug targeting androgen receptor for androgenetic alopecia therapy
Zhang et al. [34]	2022	Machine learning guided discovery of superoxide dismutase nanozymes for androgenetic alopecia

Table 2 continued

Author	Year	Title
Liu et al. [35]	2025	Performance exploration of multi-gene panels of alopecia areata susceptibility and drug-binding targets
Robotics		
Avram et al. [36]	2014	Robotic follicular unit extraction in hair transplantation
Rose et al. [37]	2014	Robotic hair restoration
Zhu et al. [38]	2024	A comparative study on the application of robotic hair restoration technology versus traditional follicular unit excision in male androgenetic alopecia
Other applications		
Li et al. [39]	2021	Research on mental stress recognition of depressive disorders in patients with androgenic alopecia based on machine learning and fuzzy K-means clustering
Jueng et al. [40]	2022	Using artificial intelligence to analyze publicly available social media posts to understand patient perspectives toward specific treatments of alopecia areata
David et al. [41]	2023	Online attitudes about the first approved systemic treatment for alopecia areata: a sentiment analysis of Reddit posts
Lim et al. [42]	2024	Exploring the potential of DALL-E 2 in pediatric dermatology: a critical analysis
Chang et al. [43]	2024	Comparison of large language models in generating patient handouts for the dermatology clinic: a blinded study
Yigen et al. [44]	2025	Artificial intelligence in patient education for androgenetic alopecia: a comparative study of ChatGPT, Gemini, and Deepseek R1

for assessing donor capacity and estimating the severity of hair loss [9]. Importantly, these systems are designed for quantitative assessment rather than etiological diagnosis, and their current clinical utility lies mainly in standardized measurement and longitudinal monitoring.

Optical coherence tomography (OCT) has also been explored as a noninvasive imaging modality for in vivo hair and follicle quantification [11]. OCT exploits the interference properties of infrared light to generate high-resolution, cross-sectional images of tissue, allowing three-dimensional analysis of hair follicles and surrounding structures. A key advantage of OCT is that it does not require hair trimming or dyeing, enabling a fully nontraceable assessment.

These systems can be utilized to objectively assess both the progression of alopecia and the response to treatment, while substantially reducing the time and effort required compared with manual evaluation. For example, HairMetrix has been used to evaluate the response to platelet-rich plasma (PRP) treatment in women with AGA [12], while another software, TrichoLAB®, has been applied to assess PRP response in men with AGA [13]. Moreover, although not originally designed for this purpose, ChatGPT has been used to monitor treatment response to minoxidil-dutasteride-copper peptide tattooing in AGA [14]. This evaluation was based on clinical photographs rather than trichoscopic images and was compared with dermatologists' evaluations, demonstrating close alignment.

Beyond AGA, ML has been applied to quantify the extent of hair loss in alopecia areata (AA), particularly for estimating the Severity of Alopecia Tool (SALT) score [15, 16]. In these studies,

images were obtained from four standardized views (left, right, top, and back) for each subject. The SALT score remains the most widely used method for assessing disease severity in AA; however, its application is inherently subjective and prone to interobserver variability. Accurate estimation of scalp involvement requires visual approximation of hair loss across predefined scalp regions, which may lead to inconsistent results, particularly in longitudinal follow-up or multicenter studies.

AI-based image analysis offers a promising solution by enabling automated scalp segmentation and quantitative estimation of hair loss from standardized clinical photographs. These systems can consistently assess the extent of alopecia, reduce observer-dependent variability, and facilitate objective monitoring of disease progression and treatment response. Such tools are particularly valuable in clinical trials and routine follow-up, where reproducible outcome measures are essential.

DIAGNOSIS

Regarding diagnostic applications, the development of AI in hair disorders remains limited compared with other areas of dermatology. At present, no AI-based tools offer reliable, real-world applicability for the comprehensive diagnosis of alopecias or for accurately distinguishing among the various alopecia subtypes. Nonetheless, several models have been proposed to specifically recognize AA as a distinct clinical entity.

One study proposed a two-step framework for the diagnosis and staging of AA. In the first step, the model was designed to discriminate AA from other scalp disorders and from healthy controls. In the second step, the AA dataset was subdivided into active, inactive, and regrowth categories, allowing the model to determine the level of disease activity [17].

Another work described a classification framework to differentiate healthy hair from AA using hair images. The authors relate that previous research in this field had relied on dermoscopic and scalp images [18].

PERSONALIZED MEDICINE

Biomarkers

Biomarkers play a central role in the development of personalized medicine, providing objective indicators of disease activity, prognosis, and therapeutic response. Traditionally, biomarker discovery in hair disorders has relied on hypothesis-driven approaches and limited datasets, often focusing on isolated molecular pathways. However, hair diseases are biologically complex and multifactorial, involving genetic, immunological, metabolic, and environmental factors.

AI enables a data-driven approach to biomarker discovery by integrating and analyzing large-scale, high-dimensional datasets, including transcriptomic, proteomic, microbiome, and clinical data. ML algorithms can identify complex, nonlinear relationships and biomarker signatures that may not be apparent through conventional statistical methods. In hair disorders, this approach has facilitated the identification of novel molecular pathways and potential therapeutic targets, contributing to a more comprehensive understanding of disease heterogeneity and treatment response.

One study leveraged multi-kingdom microbial signatures together with ML algorithms to construct a microbial index of scalp health (MiSCH) capable of detecting AGA and stratifying its severity [19].

In the context of AA, integrated ML analyses combined with immune-cell infiltration profiling have identified ferroptosis-related genes (*SLC40A1*, *LCN2*, *CREB5*, and *SLC7A11*) as promising biomarkers [20]. The findings suggest a potential role of ferroptosis-related pathways in the progression of AA. This not only deepens the understanding of the molecular mechanisms underlying the disease but also highlights new diagnostic and therapeutic possibilities.

Additional studies using skin samples from patients with AA have identified further candidates for evaluating disease risk, severity, and progression [21]. Analyses of large sets of genes, combined with ML models, have made it possible to distinguish genes closely linked to AA. Among these, *CHAC1* has shown particular

promise for assessing disease severity and for helping to differentiate AA from other types of alopecia [22].

More recently, ML approaches integrating transcriptomic data from multiple public datasets have enabled the identification of predictive gene expression signatures associated with AA, supporting early disease prediction and molecular stratification. Importantly, these models have combined robust predictive performance with interpretability analyses, providing insights into key biological pathways involved in disease onset, including immune- and stress-related mechanisms [23].

Further research has focused on finding markers that can predict progression to more extensive forms of AA, such as alopecia totalis or alopecia universalis. Several genes (*CD28*, *HOXC13*, *KRTAP1-3*, and *GPRC5D*) have been highlighted as key indicators of this progression [24]. Furthermore, ML assisted screening of immune monitoring genes has also identified *LGR5*, *SHISA2*, *HOXC13*, and *S100A3* as potential early markers of severe AA, with *HOXC13* showing particularly strong performance [25].

Progress has also been made toward more personalized treatment strategies. One study described a computational model capable of predicting which patient–drug combinations are most likely to produce a favorable therapeutic response in AA. This was achieved by integrating information on drug mechanisms of action and gene-regulatory networks with baseline patient data prior to beginning treatment [26].

Beyond predictive modeling of treatment response, recent advances further emphasize the role of AI in enabling personalized therapeutic strategies. Emerging cell-free approaches, such as exosome-based therapies, illustrate the growing biological complexity of novel treatments, characterized by heterogeneity in molecular cargo and biological activity [27].

Artificial intelligence-based methods may help manage this complexity by integrating large-scale biological data to support biomarker discovery, patient stratification, and therapeutic optimization. Although still at an early stage in hair disorders, these approaches highlight a broader shift toward data-driven precision medicine.

Clinical Markers

AI has also been explored as a tool to identify clinical predictors of treatment response. One study focused on predicting therapeutic outcomes in patients with female pattern hair loss on the basis of clinical features [28]. It showed that age at onset and duration of hair loss were important predictors of treatment response.

In AA, a model trained on trichoscopic images to predict treatment response correctly identified nonresponders in 80% of cases and responders in 81% [29]. These findings suggest that AI-based evaluation of trichoscopic patterns may be useful for predicting disease course and treatment response in AA.

ML methods have also been used to predict progression in scarring alopecias. In lichen planopilaris, a multidimensional model incorporating trichoscopy, vitamin D levels, and diagnostic timelines showed that trichoscopic severity, vitamin D deficiency, and delayed diagnosis are key, and potentially modifiable, predictors of disease progression [30].

A different direction has been the use of AI for personalizing nonmedicated treatment plans. In a 24-week study involving women with self-reported hair thinning, an AI-driven platform generated individualized treatment kits on the basis of scalp images and questionnaire data. Treatments included various combinations of topical serums, shampoos, oral supplements, and marine collagen peptides. Significant improvements were observed in hair growth, coverage, thickness and shedding, with high patient satisfaction and no reported adverse events [31].

DRUG DEVELOPMENT

AI is also beginning to influence drug discovery in the field of hair disorders, supporting the identification of new therapeutic targets and accelerating the early phases of compound development.

Regarding AGA, one study investigated nonsteroidal inhibitors of human steroid 5α-reductase type I (SRD5A1), a key enzyme

involved in the condition. Because no crystal structure exists for human SRD5A1, and current steroidal inhibitors such as finasteride and dutasteride may cause adverse effects, alternative compounds were explored. They synthesized several hydroxycinnamate derivatives and identified ferulate molecules with measurable inhibitory activity, low cytotoxicity, and favorable predicted properties. These findings indicate that such derivatives could represent nonsteroidal candidates for future anti-androgen therapies [32].

Another study explored the use of AI-based protein design to develop a new therapeutic option for AGA. In this work, advanced AI tools were used to design a peptide capable of binding to the androgen receptor and promoting its degradation through a proteolysis targeting chimera (PROTAC) mechanism. The candidate peptide was further refined using computational modeling and experimentally confirmed to bind the intended targets. To enable practical delivery, the peptide was incorporated into a microneedle patch for transdermal administration [33].

Additional work has examined superoxide dismutase (SOD) nanozymes as a potential treatment for AGA. As oxidative stress contributes to follicular damage, a manganese-based compound ($MnPS_3$) with strong SOD-like activity was identified. To allow efficient delivery to the hair follicle, the compound was incorporated into a dissolvable microneedle patch. In a mouse model, this system promoted hair regrowth more effectively than topical minoxidil, even when applied less frequently, and showed good biocompatibility [34].

In AA, AI has also been applied to identify new potential therapeutic targets. By analyzing publicly available gene-expression datasets, ML models highlighted groups of immune-related genes, particularly those associated with T cell and NK-cell activity, as contributors to follicular damage. These key genes were then used for computer-based drug screening, which identified several small molecules with predicted stable binding to their encoded proteins. Such compounds may represent candidates for future targeted therapies in AA [35].

ROBOTICS

Robotic systems represent another area in which AI has begun to influence hair disorders, particularly in the field of hair transplantation. The first Food and Drug (FDA)-approved device, the ARTAS® robotic hair transplantation system, received authorization in 2011. This platform integrates several components, including a multiaxis mechanical arm, a double-needle harvesting system, high-resolution imaging, and an AI-based follicular unit (FU) recognition module, to automate the selection and extraction of FUs while aiming to minimize follicular damage [36, 37].

Since its introduction, ARTAS has attracted interest among hair-transplant surgeons owing to its standardized operation and relatively short learning curve. Clinical evaluations have compared ARTAS with conventional follicular unit excision (FUE) techniques. In a study involving 13 male patients with androgenetic alopecia, donor areas were divided so that each patient received ARTAS-assisted harvesting on one side and manual FUE on the other. Performance metrics, including yield rate, transection rate, and discard rate, were assessed for both methods. The ARTAS system demonstrated a slightly lower yield rate and a significantly higher discard rate than manual FUE, whereas transection rates were comparable between the two techniques. No complications were reported, and patient satisfaction with the clinical outcome was similar across both methods. Overall, current iterations of robotic hair-transplant technology appear to be safe and effective for use in AGA surgery [38].

OTHER APPLICATIONS

AI applications in hair disorders are broad and heterogeneous, and several emerging uses do not fit neatly into the previous sections. For this reason, a number of additional applications are highlighted to illustrate how AI is being used in complementary domains, including emotional assessment, patient-reported outcomes, and clinical communication.

In one study using electroencephalography recordings, ML-based feature selection methods were applied to identify depressive symptoms in patients with AGA. The optimized model demonstrated higher classification accuracy than alternative approaches and was also able to monitor symptom improvement following psychological interventions [39].

Natural language processing techniques have been used to analyze large-scale datasets of patient-generated content. One study examined more than 100,000 social media posts related to AA across multiple platforms. ML-based sentiment analysis was applied to extract patient-reported perceptions of treatment effectiveness and emotional tone. While therapies such as Janus kinase inhibitors, topical minoxidil, and oral corticosteroids were frequently associated with perceived improvement, a substantial proportion of posts continued to express negative emotional sentiment, underscoring the significant psychological burden of AA [40].

Another analysis focused on public attitudes toward baricitinib, the first approved systemic treatment for AA. Using sentiment analysis of Reddit posts, the study demonstrated a significantly higher likelihood of positive sentiment in posts referencing baricitinib compared with those that did not, both before and after regulatory approval. These findings illustrate how AI-based analysis of patient-generated data can provide insights into real-world expectations and perceptions surrounding new therapies [41].

Text-to-image generative models have also been explored for educational purposes in dermatology. In a study evaluating the performance of DALL·E 2 across 12 pediatric dermatologic conditions, including alopecia areata, results varied considerably. While images of acne were often realistic, those generated for alopecia areata frequently lacked defining clinical features, such as well-demarcated patches of hair loss, and occasionally appeared cartoon-like rather than photorealistic. These findings indicate that current text-to-image models are not yet reliable for producing clinically accurate educational images of hair disorders [42].

LLMs have also been evaluated for their ability to generate patient-oriented educational materials in dermatology and hair disorders. Comparative studies in which dermatologists reviewed patient handouts generated by models such as ChatGPT, BingAI, and Bard have assessed their quality, readability, and clinical usefulness using validated instruments. Overall, ChatGPT has consistently achieved the highest ratings for content quality and usability, while ChatGPT and BingAI have produced more understandable explanations than other models, particularly in conditions related to alopecia [43].

More recent evaluations focusing specifically on AGA have confirmed that certain LLMs are capable of delivering accurate, readable, and patient-friendly information regarding disease mechanisms and treatment options [44]. However, these studies have also highlighted substantial variability between models, as well as the potential for incomplete or oversimplified information. Taken together, these findings suggest that LLMs may serve as supportive tools for patient education and clinical communication, provided that their outputs are reviewed and contextualized by clinicians, as expert verification remains essential to ensure accuracy and patient safety [43, 44].

LIMITATIONS AND ETHICAL CONSIDERATIONS

Despite the growing interest in AI-based applications for hair disorders, several limitations currently restrict their routine clinical implementation. Most available studies are exploratory, retrospective, or based on limited datasets, which may affect generalizability and reproducibility across clinical settings [45, 46]. In addition, heterogeneity in data acquisition and annotation remains a major barrier to external validation [47].

Data bias represents a further concern, as many AI models are trained on datasets that do not adequately reflect the diversity of real-world patient populations. This may result in reduced performance and unintended inequities when such tools are applied beyond their original development cohorts [48].

Interpretability is another critical issue, particularly for complex DL models. Limited transparency in algorithmic decision-making may hinder clinical trust and adoption, emphasizing the need for explainable AI approaches in high-stakes medical applications [49].

Ethical considerations are especially relevant for applications involving patient-generated data and LLMs. Variability in output quality and the potential dissemination of inaccurate information highlight the importance of physician oversight and regulatory guidance [50]. Overall, AI tools should be regarded as adjuncts rather than replacements for clinical expertise, requiring appropriate validation and supervision to ensure safe integration into trichological practice.

CONCLUSIONS

AI is reshaping multiple areas of hair disorders, including quantitative assessment, diagnostic support, personalized medicine, drug discovery, robotics, analysis of patient-reported outcomes, and clinical communication. Current tools show promising potential to complement clinical decision-making and improve efficiency, but they are not yet substitutes for expert evaluation.

ACKNOWLEDGMENTS

We thank the participants of the study.

Medical Writing/Editorial Assistance. ChatGPT was used for an English language check.

Author Contributions. David Saceda-Corralo conceived the idea for the article. Alba Lecumberri performed the literature search, conducted the data analysis, and drafted the manuscript. All authors: Alba Lecumberri, Sergio Vañó-Galván, David Saceda-Corralo, and Daniel Asz-Sigall critically reviewed the manuscript for important intellectual content and approved the final version.

Funding. No funding or sponsorship was received for this study or publication of this article.

Data Availability. Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

Declarations

Conflicts of Interest. David Saceda-Corralo is an editorial board member of *Dermatology and Therapy*. David Saceda-Corralo was not involved in the selection of peer reviewers for the manuscript nor any of the subsequent editorial decisions. Alba Lecumberri, Sergio Vañó-Galván, and Daniel Asz-Sigall have nothing to disclose.

Ethical approval. This article is based on previously conducted studies and does not contain any new studies with human participants or animals performed by any of the authors.

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